

Component Budget Report

IoT Health & Environmental Monitoring System (Pico 2W Rust Edition)

Handled by Kariyawasam L.L.N | Currency: Sri Lankan Rupees (LKR) | July 2026

1. System Overview

This budget covers the hardware components shown in the system architecture diagram: a Raspberry Pi Pico 2W acting as the main processing engine, a round TFT status display, four environmental/health sensors (MAX30100, PMS7003, SGP30, SHTC1), and a 4G/GSM shield that relays sensor data to a HiveMQ broker over MQTT.

2. Core Component Pricing

Prices were collected from two Sri Lankan electronics retailers per item where stock was available. The lowest confirmed price is used for the project total.

Raspberry Pi Pico 2 W	RP2350 MCU, Wi-Fi/BLE main processing engine	Rs. 3,250.00	—	Rs. 3,250.00
		<i>Tecnock.lk</i>	<i>(no second confirmed local retailer)</i>	
Round TFT Display	1.28" GC9A01, 240x240 round SPI LCD status display	Rs. 1,450.00	Rs. 2,190.00	Rs. 1,450.00
		<i>Tronic.lk</i>	<i>Alphatronic.lk</i>	
4G/GSM Shield	A7670C / SIM7600-class Cat-1 4G module internet uplink	Rs. 6,500.00	Rs. 7,500.00	Rs. 6,500.00
		<i>Import estimate (Daraz.lk resellers)</i>	<i>Import estimate (AliExpress + duty, converted)</i>	<i>(indicative only)</i>
MAX30100 Sensor	Pulse oximeter SpO2 & heart-rate (BPM)	Rs. 420.00	Rs. 490.00	Rs. 420.00
		<i>Tronic.lk</i>	<i>Alphatronic.lk</i>	
PMS7003 Sensor	Laser PM2.5 / PM10 particulate sensor	Rs. 5,180.00	Rs. 5,950.00	Rs. 5,180.00
		<i>Tronic.lk</i>	<i>Duino.lk</i>	
SGP30 Sensor	TVOC / eCO2 air-quality sensor	Rs. 2,480.00	—	Rs. 2,480.00
		<i>Tronic.lk</i>	<i>(no second confirmed local retailer)</i>	
SHTC1 Sensor	Temperature & humidity sensor	Rs. 1,650.00	—	Rs. 1,650.00
		<i>Tronic.lk</i>	<i>(no second confirmed local retailer)</i>	
TOTAL (core components)	Rs. 20,930.00	Rs. 16,130.00 *	Rs. 20,930.00	

* Shop 2 total excludes items with no second confirmed local retailer.

3. Sourcing Notes

- Raspberry Pi Pico 2W: only Tecnock.lk had it explicitly listed at the time of research (Rs. 3,250, discounted from Rs. 4,250); it was shown out of stock, so lead time should be confirmed before ordering.

- 4G/GSM Shield: no Sri Lankan retailer was found stocking an A7670C/SIM7600-class 4G shield locally. The figures shown are indicative import costs via Daraz.lk resellers and AliExpress (converted to LKR, order-in cost) get a direct quote before finalising the budget.
- SGP30 and SHTC1: confirmed only at Tronic.lk; no second Sri Lankan retailer stock was found for direct comparison.
- All other components (round display, MAX30100, PMS7003) were confirmed at two independent retailers, listed above.

4. Accessories & Consumables

Typical supporting items needed to assemble and power the device. These are ballpark ranges rather than two-shop comparisons, since exact parts (cable length, battery capacity, box size) vary by build.

Breadboard / prototyping PCB	Rs. 350.00	Rs. 550.00
Jumper wires (M-M / M-F / F-F set)	Rs. 250.00	Rs. 400.00
Micro-USB / USB-C cable	Rs. 250.00	Rs. 400.00
Enclosure / project box	Rs. 800.00	Rs. 1,500.00
18650 Li-ion battery + charging module (TP4056)	Rs. 950.00	Rs. 1,400.00
4G-capable micro SIM (data plan, monthly)	Rs. 500.00	Rs. 1,000.00
Soldering & misc consumables (allowance)	Rs. 500.00	Rs. 800.00
SUBTOTAL (accessories)	Rs. 3,600.00	Rs. 6,050.00

5. Estimated Project Total

Core components (best price per item)	Rs. 20,930.00
Accessories & consumables (average of range)	Rs. 4,825.00
ESTIMATED PROJECT TOTAL	Rs. 25,755.00

This total assumes one unit of each component and does not include a recurring 4G data plan beyond the SIM allowance listed above, or shipping/import duty on the GSM shield if sourced from overseas.